Dimensionality Reduction Technique

Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis (LDA) is a statistical technique used for classification and dimensionality reduction. It aims to find a linear combination of features that maximises the separation between classes in a dataset. LDA is widely used in various fields, including machine learning, pattern recognition, and statistics.

The main objective of LDA is to find a lower-dimensional space in which the data points from different classes are well-separated while keeping the data points from the same class close together. It operates under the assumption that the data follows a multivariate normal distribution and that the classes have equal covariance matrices.

Let's break down the key steps in Linear Discriminant Analysis:

1. Step 1: Compute Class Means:
Calculate the mean vector for each class by taking the average of the feature vectors belonging to that class. The mean vector represents the centroid of the data points within each class.

2. Step 2: Compute Within-Class Scatter Matrix (Within-Class Variance):
The within-class scatter matrix, denoted as \(S_W\), measures the spread of data within each class. It is calculated as the sum of the covariance matrices of individual classes weighted by the number of samples in each class. Mathematically, for each class \(i\), we compute:

$$S_i = \sum_{x \text{ in class } i} (x - \text{mean of class } i)(x - \text{mean of class } i)^T$$

and then combine them to get the within-class scatter matrix Sw.

3. Step 3: Compute Between-Class Scatter Matrix (Between-Class Variance):
The between-class scatter matrix, denoted as \(S_B\), measures the spread of the class means. It is calculated as the sum of the outer product of the difference between class means and the overall mean. Mathematically:

$$S_B = \sum_i N_i \cdot (\text{mean of class } i - \text{overall mean}) \cdot (\text{mean of class } i - \text{overall mean})^T$$

where **Ni** is the number of samples in class **i**.

4. Step 4: Solve the Generalized Eigenvalue Problem:
To find the optimal linear discriminants, we need to solve the generalised eigenvalue problem:

$$S_W^{-1} S_B \mathbf{w} = \lambda \mathbf{w}$$,

where **W** is the eigenvector, and **lambda** is the eigenvalue. We want the eigenvectors corresponding to the largest eigenvalues, as they represent the directions in the feature space that maximise class separation.

5. Step 5: Reduce Dimensionality:
Select the top **k** eigenvectors corresponding to the largest eigenvalues, where **k** is the desired lower dimensionality for the reduced feature space. These eigenvectors form a transformation matrix **W**. Project the original data onto this lower-dimensional space using **W**, resulting in the reduced feature vectors.

6. Step 6: Classification:
The reduced feature vectors can be used as input for any classification algorithm (e.g., logistic regression, k-nearest neighbors) to perform the final classification task.

LDA is particularly useful when you have a dataset with multiple classes and want to reduce the dimensionality while preserving the discriminative information. It helps in visualizing and understanding data, as well as improving the performance of classification models, especially when the number of features is large compared to the number of samples.

Mathematically -

As its primary goal is to find the linear combination of features that best separates classes in the data.

Suppose we have a dataset with N samples and D features, where each sample belongs to one of K classes. The goal of LDA is to project this D-dimensional data onto a lower-dimensional space while maximizing the separation between classes.

Let's define the following terms:

- X is the NXD matrix representing the dataset, where each row corresponds to a sample, and each column represents a feature.
- y is the N-dimensional vector representing the class labels for each sample.

Here are the main steps of LDA:

1. Compute the mean vectors of each class:

$$\mu_k = \frac{1}{N_k} \sum_{i=1}^{N} x_i \text{ for } k = 1, 2, \ldots, K$$

where **Nk** is the number of samples in class **k**, and **xi** is the **i-th** row (sample) of **X**.

2. Compute the scatter matrices:
   - Within-class scatter matrix:

$$S_W = \sum_{k=1}^{K} \sum_{i=1}^{N} (x_i - \mu_k)(x_i - \mu_k)^T$$

- Between-class scatter matrix:

$$S_B = \sum_{k=1}^{K} N_k(\mu_k - \mu)(\mu_k - \mu)^T$$

where **mu** is the overall mean of the data:

$$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$$

3. Solve the generalized eigenvalue problem $S_W^{-1} S_B \mathbf{w} = \lambda \mathbf{w}$ to find the linear discriminants. The vectors **w** with the largest eigenvalues are the directions along which the data should be projected to achieve the best class separation.

4. Select the top L linear discriminants corresponding to the L largest eigenvalues. L is the desired lower dimensionality for the reduced feature space.

5. Project the data onto the new feature space using the selected linear discriminants:

$$Y = X\mathbf{W}$$

where Y is the **NXL** matrix representing the reduced feature space, and **W** is the **DXL** matrix containing the top L eigenvectors as columns.

LDA can be used for classification tasks by training a classifier (e.g., logistic regression or k-nearest neighbours) on the reduced feature space Y.

Note: LDA assumes that the data follows a multivariate normal distribution, and the classes have equal covariance matrices. If these assumptions are not met, LDA may not perform optimally. Additionally, when the number of classes is greater than D-1, the scatter matrices may be rank-deficient, requiring regularisation techniques.

In practice, many machine learning libraries provide implementations of LDA, so you don't need to code it from scratch.

LDA (Linear Discriminant Analysis) Vs PCA (Principal Component Analysis)

LDA (Linear Discriminant Analysis) and PCA (Principal Component Analysis) are both dimensionality reduction techniques used in machine learning and data analysis. However, they serve different purposes and operate under different assumptions. Here are the main differences between LDA and PCA:

1. Objective:
- LDA: The primary goal of LDA is to find a lower-dimensional space that maximizes the separation between classes in a dataset. It focuses on preserving the class-specific information to enhance classification performance.

- PCA: PCA aims to find the lower-dimensional space that captures the maximum variance in the data. It does not consider the class labels and seeks to represent the data in terms of orthogonal components (principal components).

2. Supervised vs. Unsupervised:
- LDA: LDA is a supervised technique that requires labeled data (class labels) during training. It leverages the class information to find the optimal feature subspace.

- PCA: PCA is an unsupervised technique and does not require class labels. It operates solely based on the covariance matrix of the data.

3. Application:
- LDA: LDA is mainly used for classification tasks, where the goal is to improve the performance of a classifier by reducing the feature space while retaining the class-discriminative information.

- PCA: PCA is often used for data compression, visualization, denoising, and preprocessing steps to remove redundant or less informative dimensions.

4. Data Assumptions:
- LDA: LDA assumes that the data follows a multivariate normal distribution and that the classes have equal covariance matrices.

- PCA: PCA assumes that the data is centered (zero mean) and that the principal components (eigenvectors) are orthogonal.

5. Output Dimensions:
- LDA: In LDA, the number of output dimensions is typically limited to \(K-1\), where \(K\) is the number of classes. It aims to create a feature space that can best separate the classes.

- PCA: PCA does not depend on class labels, and the number of output dimensions is usually determined by the user or based on the desired level of variance explained.

6. Use Cases:
- LDA: LDA is useful when dealing with classification tasks, such as face recognition, sentiment analysis, or speech recognition, where the class information is essential for distinguishing between data points.

- PCA: PCA is often used for data visualisation, feature extraction, noise reduction, and data preprocessing before applying other machine learning algorithms.

In summary, while both LDA and PCA are dimensionality reduction techniques, LDA focuses on class separation for classification tasks with labelled data, while PCA aims to capture the variance and reduce redundancy in the data without using class labels. The choice between LDA and PCA depends on the specific problem and whether the class information is available or relevant for the task at hand.